# COMSATS University Islamabad (CUI)

# Project Proposal

**for**

# LEADS

(Lead generation Engine for Automated Discovery System)

Version 1.0

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# Table of Contents

Abstract ..... 1
1. Introduction ..... 2
2. Problem Statement ..... 2
3. Problem Solution/Objectives of the Proposed System ..... 3
   3.1 Objectives ..... 3
4. Related System Analysis/Literature Review ..... 4
5. Vision Statement ..... 5
6. Scope ..... 6
7. Modules ..... 6
8. System Limitations/Constraints ..... 12
9. Data Gathering Approach ..... 13
10. Project Contribution ..... 13
11. Relevance to Course Modules ..... 14
12. Tools and Technologies ..... 15
13. Project Stakeholders and Roles ..... 17
14. Module based Work Division ..... 17
15. WBS and Gantt Chart ..... 18
16. Mockups ..... 23
17. References ..... 26
18. Plagiarism Report ..... 27

**Project Category:**

- Web Application
- Problem Solving and Artificial Intelligence

# Abstract

LEADS is a web-based real estate intelligence and outreach platform built for the Pakistani property market. The platform operates as an active business tool rather than a passive listing board. Its primary function is to help real estate professionals and investors identify high-demand property categories using AI-driven market trend analysis, find matching seller leads through targeted on-demand scraping of four competing platforms, engage those sellers through structured WhatsApp outreach, collect property details through category-specific digital forms, analyse the collected data using large language models to generate valuations and AI-enhanced listing descriptions and manage the resulting warm leads through a structured campaign and seller dashboard.

Unlike every major Pakistani property platform today, including Zameen.com, OLX Pakistan, Graana.com, Ilaan.com which function as passive listing boards where sellers post and wait indefinitely, LEADS actively works the market on behalf of its users. The system scrapes publicly available listing data from all four platforms continuously, builds a market intelligence layer that computes listing velocity, average days on market and price movement per category and city and uses this computed intelligence to surface what is currently trending. Users then trigger on-demand targeted scrapers with specific filters, review AI-generated seller outreach messages before sending, collect structured property data through WhatsApp-delivered forms and receive warm lead notifications when sellers respond. The platform also includes a listing management system where confirmed properties can be published for buyers to browse and discover. To the best of our knowledge, LEADS is the first platform in Pakistan to combine real-time market intelligence computed from multi-platform scraped data, AI-powered seller outreach drafting, structured property data collection, warm lead detection via WhatsApp webhook and an integrated listing management system in one connected pipeline. It does not just publish a listing and walk away it actively works for both the seller and the buyer from the moment a listing goes live.

# 1. Introduction

This document presents the complete project scope for LEADS, a web-based real estate platform developed for the Pakistani market as a Final Year Project submission and its purpose is to formally define the system's background, objectives, modules, technical decisions and development plan in a structured and reviewable format. Pakistan's real estate sector is one of the largest industries in the country yet the technology supporting buyers and sellers remains stuck in a passive model where a seller posts a listing and waits indefinitely for a buyer to discover it, with no active outreach, no intelligent matching and no quality control. Platforms like Zameen.com, OLX Pakistan, Graana.com, Ilaan.com and all follow this same pattern and none of them verify user identities, enforce listing quality or do anything to actively connect the right buyer to the right seller. This gap in the market is what LEADS is built to address, by embedding automated lead generation, AI-assisted listing review, identity verification, lead scoring and direct buyer-seller communication into a single platform purpose-built for Pakistani real estate. The document begins by defining the problem in detail and presenting the proposed solution and system objectives, followed by a review of existing platforms and how LEADS differs from each of them. It then defines the full scope of the system, the twelve modules it is composed of and the specific functional requirements each module must deliver. The tools and technologies selected for development are listed and justified, the system's limitations and constraints are documented and the data gathering approach used to validate the problem is described. Project stakeholders and their roles are identified, the work is divided between team members and a work breakdown structure and project timeline are provided. Mockups of the key screens across all three user portals are included to illustrate the intended user experience. The document concludes with a references section and a plagiarism report.

# 2. Problem Statement

Real estate agents and sellers in Pakistan currently have no platform that actively helps them to find serious buyers, they are completely dependent on buyers finding them first. On every major property platform in Pakistan, including Zameen.com, OLX Pakistan, Graana.com and Ilaan.com, the process is the same, a seller posts a listing and keep waiting, sometimes for weeks or even months, without receiving a single genuine inquiry. These platforms do nothing after a listing goes live as there is no outreach, no follow-up and no system that matches the listing to buyers who are looking for that type of property. Sellers also have no way of knowing whether their asking price is too high or too low compared to what similar properties are currently listed for on other platforms. On top of this, there is no identity verification on most platforms that means anyone can post a list anonymously and fake listings are very common. OLX Pakistan, for example, publishes listings immediately with no review at all, making it easy for unverified sellers to mislead buyers. Even the platforms that review listings before publishing, the quality of descriptions is often poor, with missing details, misleading categories and prices that do not reflect the actual market trends. Buyers waste a significant amount of time visiting properties that do not match their need. Sellers with legitimate properties get buried under low-quality and fake listings, making it much harder for the right buyers to find them. Real estate agents face the same problems and spend a large part of their working day manually searching for leads, sending messages one by one and posting listings across multiple platforms with no automation or intelligence to support them.

# 3. Problem Solution/Objectives of the Proposed System

LEADS replaces the passive, manual and fragmented process of Pakistani real estate with a connected, AI-powered and automated pipeline. The system operates across five distinct stages: market intelligence, targeted scraping, seller outreach and data collection, buyer outreach and warm lead detection and listing management. Each stage is implemented as one or more dedicated modules with clearly defined inputs, outputs and technical implementation. In the first stage, LEADS computes what is trending in the property market by continuously analysing scraped listing data from four platforms. For each category and city combination, the system tracks listing velocity, average days on market and price movement. These three signals are combined into a demand score per category per city and passed to a large language model to generate a human-readable trend recommendation. This gives users, including those with no prior real estate experience, a data-driven starting point for their investment decisions. In the second stage, once a user selects a category, city and price range, LEADS launches a targeted multi-agent scraping run. Rather than one scraper trying to collect everything, the system launches one Celery worker per platform, each assigned a different proxy IP, each collecting a fixed maximum of records in parallel. The four workers run simultaneously, their results are cleaned, deduplicated and merged and the top matching listings are surfaced to the user for review. In the third stage, the user selects a listing they want to pursue. The system drafts a personalised WhatsApp outreach message and a terms-and-conditions summary for the user to review and approve before sending. When the seller responds, the system sends them a category-specific property data collection form via WhatsApp. The seller fills the form once, providing all required property details, which the system then passes to a large language model to generate a fair market valuation, identify the property's hot selling features, flag any issues, confirm the correct category and produce an AI-enhanced listing description. In the fourth stage, the system uses the confirmed property details to run a second targeted scraping pass to identify potential buyers: people who are active on the same platforms in the same category and price range. The system drafts a WhatsApp campaign for user review and approval, sends it and monitors all replies through a WhatsApp Cloud API webhook. Any reply, including a thumbs up or a short acknowledgement, is classified as a warm lead, routed to the seller dashboard and triggers an automated follow-up message draft for user review. After the follow-up is sent, the system's automated role ends and the agent takes over manually. In the fifth stage, once a property is confirmed and its AI-enhanced listing is ready, it is published in LEADS' integrated listing management system where buyers can browse, filter and contact sellers directly. Any buyer inquiry from this system is also surfaced on the seller dashboard.

## 3.1 Objectives

The following objectives define the specific goals that LEADS is built to achieve in order to solve the problems identified above.

**BO-1:** Compute market demand scores per property category and city from continuously scraped listing data by tracking listing velocity, average days on market and price movement and surface this intelligence to users as AI-generated trend recommendations.

**BO-2:** Run on-demand, filter-driven, multi-agent scraping across five platforms simultaneously with each agent on a separate proxy IP and a configurable maximum record count per agent, so targeted data collection completes in minutes rather than hours.

**BO-3:** Draft personalised WhatsApp outreach messages and terms-and-condition summaries for user review and approval before any message is sent to a seller, ensuring no automated communication goes out without human confirmation.

**BO-4:** Deliver category-specific property data collection forms to sellers via WhatsApp, with different mandatory and optional fields per property type including house, flat, shop, plot and plaza, so all required property details are collected in a single structured submission.

**BO-5:** Analyse submitted form data using a large language model to generate a fair market valuation with minimum, maximum and recommended price, identify hot selling features, flag issues, verify the property category and produce an AI-enhanced listing description.

**BO-6:** Run a second targeted scraping pass using the confirmed property details as filters to identify potential buyers on all five platforms, draft a WhatsApp campaign for user review and approval, dispatch it and track delivery and engagement status per message.

**BO-7:** Monitor all incoming WhatsApp replies via webhook, classify each reply as hot, warm or cold based on response content, route warm and hot leads to the seller dashboard with a notification and auto-draft a follow-up message for user review and approval.

**BO-8:** Publish confirmed, AI-enhanced property listings in the integrated LEADS listing management system where verified buyers can browse, filter, save and contact sellers directly through the platform.

**BO-9:** Provide administrators with full control over scraper scheduling and health monitoring, proxy pool management, LLM API credentials, WhatsApp API configuration, scoring weights and platform-wide analytics through a dedicated admin control panel.

**BO-10:** Require all registered users to verify both their email address and phone number through time-limited OTP before accessing any platform feature, ensuring no anonymous participation.

# 4. Related System Analysis/Literature Review

The following section reviews the existing platforms and systems related to LEADS, identifies their weaknesses and explains how LEADS addresses each of those gaps through its proposed features and design.

Zameen.com is the biggest real estate portal in Pakistan and it operates like a listing board where the sellers post their properties and hope that someone from the other side contact them. It doesn't have any mechanism for generating leads, nor does it have an AI listing review or a marketing system that acts once the listing goes live. Buyers must search for listings on their own and sellers have no control over who contacts them.

OLX Pakistan is a general classifieds platform that covers many categories including property. Anyone can post a listing without identity verification or any quality review which makes fake and misleading property listings very common. There is no lead generation, no AI assistance and no matching between buyer needs and seller listings. The platform is free and open which lowers the overall quality of listings buyers see.

Graana.com is an example of a Pakistani online marketplace that provides more structured features than OLX, including agents and a listing value estimator feature. Yet, it too functions as a passive marketplace because once buyers make a search, no further help from the platform is provided to them. In fact, sellers receive absolutely no assistance even after posting their listing. There is no automated lead generation nor any listing optimization using artificial intelligence.

Ilaan.com is one of the popular Pakistani websites for buying and selling property. It has interesting features like viewing count and market valuation of listings. Even then, it is also a passive marketplace lacking intelligent listing services such as lead generation and outreach. The sellers can neither

determine which buyers are really interested in their listings, nor can they contact the interested buyers proactively. No AI-based listing review and listing marketing via social networks are available.

**Table 1 Related System Analysis with LEADS solution**

| Application Name | Weakness | LEADS Solution |
|---|---|---|
| Zameen.com | Passive listing board with no automated outreach, no lead scoring and no AI listing review. Sellers post and wait with no active support. | LEADS actively finds and contacts buyer leads and reviews every listing with AI before publication.es. |
| OLX Pakistan | No identity verification and no listing quality check. Anyone can post anonymously, making fake and misleading listings very common. | LEADS requires dual OTP verification from every user and enforces mandatory listing quality standards before any listing goes live. |
| Graana.com | Still a passive platform despite having a Wanted section. Sellers cannot access or act on buyer intent signals in any automated way. | LEADS collects and scores publicly available data including buyer intent signals and automatically contacts matched leads on the seller's behalf. |
| Ilaan.com | Has useful features like view counts and valuations but offers no automated buyer outreach, no lead scoring and no external listing marketing. | LEADS uses Ilaan's publicly available listing data as a market intelligence input and goes further by actively contacting |

# 5. Vision Statement

For real estate sellers and agents in Pakistan who struggle to find genuine buyers in a market flooded with fake inquiries, unverified users and platforms that do nothing after a listing goes live, LEADS is an AI-powered web-based real estate platform that automatically verifies every user through dual OTP authentication, reviews and improves every listing through Google Gemini before publication, collects and scores publicly available data from four competing platforms to identify high-intent buyer leads and contacts those leads through WhatsApp on the seller's behalf without any manual effort required. Unlike Zameen.com, OLX Pakistan, Graana.com, Ilaan.com and , which simply post listings and leave sellers waiting with no outreach, no matching and no quality control, our product LEADS transforms the seller's experience from passive waiting to active results putting verified, scored and contacted buyer leads directly in the seller's hands so that the right buyer finds the right property faster, with less effort and through a platform both sides can actually trust.

# 6. Scope

LEADS is a web-based real estate platform built for the Pakistani market that covers everything happening in the pre-transaction phase, from market intelligence to active lead generation and listing management. The platform serves real estate agents, investors and internal administrators, while external buyers interact with properties through a centralized listing hub where every user must first verify their identity via a two-step OTP process. The market intelligence system continuously scrapes Zameen.com, OLX Pakistan, Graana.com and Ilaan.com using Scrapy, Playwright and parallel proxy-assigned Celery workers, cleaning raw records to calculate listing velocity, days on market and price movements before sending a computed demand score to GPT-4o mini to generate actionable trend recommendations. When a target property is selected, the platform handles seller outreach via user-approved WhatsApp messages, collects structured property details through automated forms and passes the input to an LLM to generate fair market valuations, highlight hot features, correct categories and write enhanced listing descriptions. To connect with potential buyers, LEADS executes a second targeted scraping pass, drafts approved WhatsApp outreach campaigns, monitors replies through webhooks, automatically classifies positive responses as warm leads and generates follow-up drafts for agent handover. Once approved, listings publish to LEADS' verified listing management hub, enabling buyers to filter properties, save favorites, set alerts and contact sellers directly. LEADS operates strictly as a pre-transaction tool, avoiding payment processing, legal documentation or financial intermediation and launches exclusively in Pakistan in English.

# 7. Modules

**Module-01 User Authentication**

This module controls identity verification and access for all users. It is implemented as a set of Next.js API routes backed by Prisma ORM on PostgreSQL. OTPs are hashed using bcrypt before storage never stored as plain text. JWTs are issued in HttpOnly cookies on successful login. Phone OTP is delivered via WhatsApp Cloud API rather than SMS, as WhatsApp penetration in Pakistan significantly exceeds SMS reliability for this use case.

**FE1**: Accept a registration request collecting full name, email address, phone number and role selection. Validate all fields with Zod before any database operation.

**FE2**: Generate a 6-digit OTP, hash it with bcrypt, store it with a 10-minute expiry in the emailOtpHash and emailOtpExpiresAt fields of the User model and deliver it to the user's email via SendGrid.

**FE3**: Generate a second 6-digit OTP, hash and store it in phoneOtpHash and phoneOtpExpiresAt and deliver it to the user's phone via WhatsApp Cloud API after email OTP is successfully verified.

**FE4**:Activate the account only after both OTP hashes are verified within their expiry windows. Phone OTP is optional at registration but required before the user can initiate any outreach action.

**FE5**: Enforce a rate-limited OTP resend mechanism using the otpAttempts counter in the User model. Block resend after the configured maximum attempt count is reached.

**FE6**: Authenticate returning users via email and password, verify the bcrypt hash, check isEmailVerified before granting access and issue a signed JWT stored in an HttpOnly cookie.

**FE7**: Apply role-based access control at the middleware layer, routing ADMIN users to the admin panel, SELLER users to the seller pipeline and BUYER users to the listing discovery interface.

**FE8:** Support a secure password reset flow via a time-limited signed reset token delivered by email.

**FE9**:Verify that the new password does not match the existing bcrypt hash before updating.

**FE10:** Invalidate the session and clear the JWT cookie on user-initiated logout.

**Module-02 Market Intelligence Engine**

This module is the analytics brain of LEADS. It runs as a scheduled Celery Beat task that processes the continuously accumulated MongoDB collection of scraped listing records. For each property category and city combination, it computes three demand signals and passes the results to a large language model to generate a human-readable recommendation.

**FE1:** For each category and city combination present in the MongoDB scraped records, compute listing velocity: count the number of new records with a scraped_at timestamp in the last 7 days versus the 7 days before that, expressed as a percentage change.

**FE2:** Compute average days on market as a proxy for sell-through speed: for each category and city, calculate the mean age in days of all active listings (difference between scraped_at and current date), where a lower average indicates faster turnover and higher demand.

**FE3:**Compute price movement: calculate the mean price_numeric of records scraped in the last 14 days versus the prior 14-day window for each category and city and express the change as a percentage.

**FE4:** Combine the three signals into a composite demand score using configurable weights stored in the admin settings table in PostgreSQL. Store the computed demand scores per category per city in a dedicated analytics collection in MongoDB with a timestamp.

**FE5**: Pass the top-ranked category and city combinations with their demand scores, velocity figures and price movement percentages to Google Gemini 2.0 Flash via a structured prompt instructing the model to generate a concise, plain-language investment recommendation based on this data.

**FE6**: Store the LLM-generated recommendation text alongside the computed scores in MongoDB. Surface the recommendation and the underlying score breakdown on the user dashboard, refreshed after each analytics cycle.

**FE7**: Run the full analytics and recommendation pipeline on an admin-configurable Celery Beat schedule, defaulting to once every 24 hours and log job outcomes to the admin audit log.

**Module-03 Targeted Scraping Engine:**

This module performs on-demand, targeted, multi-agent scraping triggered when a user selects a property category, city and price range to pursue. Rather than running a single scraper across all pages of all platforms, the system launches one independent Celery worker per platform, each assigned a unique IP address from the proxy pool, each configured with a maximum record count. All five workers run concurrently. This architecture distributes request load across IPs and platforms, significantly reducing detection risk while completing the full data collection in a fraction of the time a sequential approach would require.

**FE1**:Accept a scraping request from the user dashboard containing category, city, minimum price and maximum price as parameters. Validate these parameters and pass them as arguments to the multi-agent launcher.

**FE2:** Launch one Celery task per platform (Zameen.com, OLX Pakistan, Graana.com, Ilaan.com) simultaneously. Each task is assigned a unique proxy IP from the Redis-backed proxy pool and a configurable maximum record count, defaulting to 2,500 records per agent.

**FE3**: Each agent runs a custom Scrapy spider with the category, city and price range passed as search parameters. Scrapy handles Zameen, OLX .

**FE4**: Playwright handles Graana and Ilaan. Each spider applies randomized request delays between 3 and 7 seconds, rotates user-agent strings from a pool of 13+ modern browser signatures and respects the max_records limit by counting items yielded and closing after the limit is reached.

**FE5**: Clean raw records through the shared cleaning pipeline: strip HTML artifacts, normalize price to integer PKR, normalize size to square feet with unit stored separately, standardize city names to a canonical list and reject any record with a null price after normalization.

**FE6**: Deduplicate across all five agents' outputs by hashing (platform, listing_id) combinations and checking against the MongoDB unique index. Use fuzzy matching on title, price and locality for cross-platform deduplication where listing IDs differ.

**FE7**: Store all cleaned, deduplicated records in MongoDB with fields: source_platform, listing_id, title, price_numeric, size_sqft, city, locality, seller_name, seller_phone, listing_url, scraped_at, listing_age_days, price_per_sqft, scraped_week and the session_id linking the records to the user's scraping request.

**FE8**: On completion of all five agents, merge results, rank by relevance to the user's filters (exact category match, price within range, recency) and surface the top 10 to 15 records on the user dashboard for review.

**FE9**: Log each agent's outcome (records fetched, records rejected, records deduplicated, errors, execution time) to the admin audit log and update the scraper health status on the admin dashboard.

### Module-04 Seller Outreach Agent:

This module handles all WhatsApp communication directed at sellers identified through the targeted scraping engine. Every message that goes out requires explicit user review and approval. No outreach is sent automatically. The module uses the WhatsApp Cloud API for dispatch and tracks delivery status per message.

**FE1**: When the user selects a listing to pursue, retrieve the seller's phone number, name and listing details from MongoDB. Use these to populate a structured prompt sent to GPT-4o mini, which returns a personalized outreach message draft and a terms-and-conditions summary including the commission structure.

**FE2**: Display the AI-drafted message and T&C summary to the user in a review interface showing the original seller data alongside the drafted content. The user can edit any part of the draft before approving.

**FE3**: On user approval, format the phone number to WhatsApp-compatible international format (adding 92 prefix, removing leading zero) and dispatch the message via WhatsApp Cloud API v21+ using the configured WABA-approved message template.

**FE4**: Log the dispatch event to PostgreSQL: seller_id, listing_id, message_type, dispatch_timestamp, delivery_status (SENT, DELIVERED, READ, FAILED). Update the delivery status in real time via WhatsApp webhook callbacks.

**FE5**: When the seller replies to the outreach message, detect the reply via the WhatsApp Cloud API webhook (M-07 handles reply classification for buyer outreach; seller replies are handled here), record the reply content and trigger M-05 to generate and send the property data collection form.

**FE6**: If message delivery fails, retry once via WhatsApp after 30 minutes and mark the lead as UNDELIVERED after the retry fails. Display the undelivered status on the seller dashboard.

### Module-05 Seller Form System:

This module generates and delivers category-specific property data collection forms to sellers who have responded to the initial outreach. Forms are delivered as links via WhatsApp. Each property type has a different set of mandatory and optional fields reflecting what is relevant for that type. The

goal is to replace the back-and-forth messaging that currently consumes most of an agent's time with a single structured submission that captures all required data at once.

**FE1**: Generate a unique form URL for each seller-listing pair. The URL is tied to the seller's phone number and the listing session ID to prevent form submissions from other parties.

**FE2**: Dispatch the form link via WhatsApp Cloud API alongside a brief explanatory message. The message is drafted by the system and reviewed by the user before sending.

**FE3**: Render the appropriate form based on property category: House/Apartment form (bedrooms, bathrooms, floors, kitchen, lounge, dining, gas, electricity, wiring condition, water supply, servant quarter, boundary wall, parking, documents available, registry status, asking price, final price); Shop/Office form (floor, area, frontage, mezzanine, storage, commercial zone, utilities, documents); Plot form (dimensions, corner or non-corner, utility connections, society, plot number, file or ground); Plaza/Building form (total floors, commercial units, residential units, lifts, utilities, rental income, documents).

**FE4**: Validate all mandatory fields on submission using Zod on the FastAPI backend. Return field-level validation errors to the form for correction before accepting the submission.

**FE5**: On successful submission, store the form data in PostgreSQL linked to the seller record and the listing session. Immediately trigger M-06 to begin LLM analysis of the submitted data.

**FE6**: Notify the user on the dashboard that the seller's form has been submitted and that AI analysis is in progress.

### Module-06 AI Property Analyzer:

This module processes the structured property data submitted through M-05 using large language model to generate actionable intelligence. It produces a market valuation, identifies selling points and issues, verifies the property category and generates an AI-enhanced listing description. All outputs are shown to the user before any further action is taken.

**FE1**: Retrieve the submitted form data for a property from PostgreSQL and combine it with the relevant market intelligence data from MongoDB: comparable listings in the same category, city and size range with their prices and listing ages.

**FE2**: Construct a structured prompt containing the full form data and market context and send it to Google Gemini 2.0 Flash or GPT-4o mini (configurable per admin settings) requesting: (a) fair market price estimate with minimum, maximum and recommended values, (b) three to five key selling features of the property, (c) any issues or weaknesses that could affect the sale, (d) the most appropriate property category based on the submitted details.

**FE3**: Construct a second prompt using the form data and the LLM's analysis output and send it to the LLM requesting an AI-enhanced listing description of minimum 150 words, written in professional English, structured to highlight the property's selling features and location, suitable for direct publication.

**FE4**: Store all LLM outputs (valuation range, selling features, issues, category recommendation, enhanced description) in PostgreSQL linked to the seller and listing records.

**FE5**: Display all LLM outputs to the user in a structured review interface. The user can accept the outputs, edit any section or request a regeneration before proceeding.

**FE6**: On user approval, mark the property as analysis-complete in the database and make it available for the next stage: buyer outreach via M-07 and listing publication via M-08.

**FE7**: Handle LLM API errors gracefully by logging the error, notifying the user with a retry option and never blocking the pipeline on API unavailability.

### Module-07 Buyer Outreach and Warm Lead Detection:

This module handles the buyer side of the outreach pipeline. It triggers a second targeted scraping pass using the confirmed property's details as filters, drafts a WhatsApp campaign for user review, dispatches it via the WhatsApp Cloud API, monitors all replies through the webhook, classifies each reply, routes warm and hot leads to the seller dashboard and auto-drafts a follow-up message for user review. After the follow-up is sent, the system's automated responsibility ends.

**FE1**: On user initiation of a buyer outreach campaign, call M-03's multi-agent scraper with the confirmed property's category, city and price range as filters, requesting records that represent active buyers (people listing properties in the same segment, indicating they are active market participants).

**FE2:** After the scraping run completes, apply the lead scoring model: score each record on buyer intent signal strength (NLP analysis of listing text using Hugging Face Transformers), budget alignment (price of their listing relative to the property's asking price), category compatibility (category match score) and recency (decay-weighted score based on listing age). Combine scores into a composite using admin-configurable weights.

**FE3**: Filter records below the minimum composite score threshold (admin-configurable, default 50/100) and archive them. Present the passing records to the user ranked by composite score with score breakdown visible.

**FE4**: Use the property details and the top-ranked lead records to generate a WhatsApp campaign message draft via GPT-4o mini. Display the draft to the user for review and editing before any send action.

**FE5**: On user approval, dispatch the campaign messages to all approved leads via WhatsApp Cloud API. Log dispatch status (SENT, DELIVERED, READ, FAILED) per lead in PostgreSQL. Display live campaign stats on the Campaign Dashboard (M-11).

**FE6**: Monitor all incoming WhatsApp replies via the WhatsApp Cloud API webhook. For each reply, identify which campaign message it corresponds to and classify it: HOT (contains 'interested', 'yes', 'tell me more', 'price?' or similar positive intent keywords), WARM (any other reply including thumbs up, 'ok', 'thanks', short acknowledgement), COLD ('not interested', 'wrong number', 'remove me').

**FE7**: For HOT and WARM classified replies, create a warm lead record in PostgreSQL linked to the lead, the campaign and the property. Send a real-time notification to the user dashboard and campaign dashboard.

**FE8**: Auto-draft a follow-up WhatsApp message for each warm lead using the property details and the content of the lead's reply. Display the draft to the user for review and approval. On approval, dispatch the follow-up via WhatsApp Cloud API. After the follow-up is sent, mark the lead as FOLLOW_UP_SENT and the system's automated involvement with this lead ends.

### Module-08 Listing Management System:

This module is the buyer-facing listing platform integrated within LEADS. When a property has been confirmed, analysed by AI and approved by the user, it is published here. Buyers browse, filter, save and contact sellers through this system. It is built on the same Next.js frontend as the rest of LEADS but serves a separate buyer audience. All buyers must complete OTP verification before accessing any listing.

**FE1**: Receive confirmed, AI-analysed property listings from M-06, combine them with the AI-enhanced description, valuation context and seller-verified details and create a fully structured listing record in PostgreSQL ready for publication.

**FE2**: Serve all active listings to authenticated buyers via server-side rendered Next.js pages, indexed by Google through Schema.org PropertyListing structured data markup on each listing detail page.

**FE3**: Provide multi-criteria filtering on the listing index page: property category, city, locality, price range, size range, transaction type (sale or rent) and publication date.

**FE4**: Display a listing summary card showing the primary image, category, city, price and a 2-sentence excerpt from the AI-enhanced description. Display a full detail page showing all property attributes, the complete AI-enhanced description, all images and an AI-generated valuation context snippet.

**FE5**: Allow authenticated buyers to save listings to a personal bookmarks collection and configure saved search preferences with email alerts when new listings matching their criteria are published.

**FE6**: Allow buyers to contact the seller through a contact request form. Route the contact request to the seller's notification feed on the seller dashboard. Log the contact event as a listing-sourced warm lead in PostgreSQL.

**FE7**: Allow administrators to publish, pause or permanently remove any listing from the admin control panel.

### Module-09 Seller Dashboard:

This module provides the primary interface for sellers and agents to monitor their active deals, lead pipeline, campaign performance, AI analysis results and revenue tracking.

**FE1:** Display a summary of all active scraping sessions, showing the top matching listings retrieved, their scores and the current outreach status for each.

**FE2**: Display all warm leads received across all campaigns, ranked by classification (HOT first, then WARM), with the original reply content, the property they are interested in and the follow-up status.

**FE3**: Display the AI analysis results for each confirmed property: valuation range, selling features, issues and the AI-enhanced listing description, with options to edit and re-submit for AI regeneration.

**FE4**: Allow the user to mark a deal as closed, enter the final agreed sale price and record the commission earned. Display cumulative revenue and deal count in a summary panel.

**FE5**: Show a 30-day activity chart covering warm leads received, campaign messages sent and deals closed.

### Module-10 Admin Control Panel:

This module gives system administrators full operational control over every component of LEADS.

**FE1**: Display a platform-wide analytics panel: total registered users by role, total scraping sessions run, total leads identified, total campaign messages sent, total warm leads detected, total deals closed, total revenue recorded and system uptime.

**FE2**: Provide a scraper management panel showing each platform scraper's status, last run time, records collected in the last 24 hours, success rate and error log. Allow manual trigger, pause and reconfiguration of any scraper.

**FE3**: Provide a proxy pool management panel showing active proxies, their current assignment status and any blocked IPs. Allow adding, removing or pausing individual proxies.

**FE4**: Provide an API credentials panel for secure storage and rotation of: Google Gemini API key, OpenAI API key, Hugging Face API key, WhatsApp Cloud API token and Phone Number ID, SendGrid API key, Google Maps API key.

**FE5**: Provide a scoring configuration panel to adjust the weights used in the composite lead score (intent, budget, category, recency) and set the minimum score threshold for outreach eligibility.

**FE6**: Provide a user management panel to search, filter by role, view, approve, suspend or soft-delete any registered user account.

**FE7:** Provide a full audit log of all admin actions, scraper job outcomes, API call events and system errors with timestamps.

### Module-11 Campaign Dashboard:

This module provides real-time visibility into the performance of every outreach campaign sent through LEADS.

**FE1**: Display a list of all campaigns (seller outreach and buyer outreach) with campaign name, property, date sent and current overall status.

**FE2**: For each campaign, display per-message stats updated in real time via WhatsApp webhook callbacks: total sent, delivered, read, replied, failed, duplicate numbers rejected.

**FE3**: Display a reply feed showing all incoming replies for a campaign with the reply content, sender phone number, classification (HOT/WARM/COLD) and timestamp.

**FE4**: Allow the user to filter campaigns by date range, property, classification outcome and delivery status.

### Module-12 User Profiling and Settings:

This module allows all users to manage their account details and preferences.

**FE1**: Allow users to update their full name, email address and phone number. Require OTP re-verification through the relevant channel if email or phone is changed.

**FE2**: Allow users to change their password through a flow requiring current password confirmation. Verify that the new password does not match the existing hash.

**FE3**: Allow buyers to configure property preferences (categories, cities, price range) used for saved search alerts.

**FE4**: Allow sellers to configure notification preferences controlling which events trigger in-platform and WhatsApp alerts.

**FE5**: Allow any user to request account deletion, confirm through a two-step confirmation flow and trigger anonymity of all personal data in compliance with the platform's data retention policy.

# 8. System Limitations/Constraints

The following section outlines the boundaries and constraints of LEADS that define what the platform does not cover in this release.

1. LEADS is a pre-transaction platform only. It does not process property payments, handle rental agreements, manage legal documents or act in any financial or legal intermediary capacity.
2. The platform is limited to Pakistan in this release. Multi-currency and multi-language support are not included.
3. The platform is in English only. AI listing enhancement and form processing work correctly for English-language inputs. Urdu submissions may not be processed accurately.
4. The scraping pipeline only collects data publicly visible without authentication. Private messages, hidden contact details and data behind login walls on any target platform are never accessed. Facebook Marketplace is excluded because it requires login to view listings.
5. The multi-agent scraping strategy requires a pool of rotating proxy IPs. Without proxy rotation, concurrent requests from the same IP will be blocked by target platforms. The quality and availability of the proxy pool directly affects scraping reliability.
6. WhatsApp Cloud API outreach requires WABA-approved message templates. Template approval can take 24 to 72 hours. Unapproved templates cannot be sent to users who have not previously messaged the LEADS WhatsApp number. This limits cold outreach to template messages only.

7. The warm lead detection webhook relies on the WhatsApp Cloud API receiving and forwarding all replies. Message delivery and reply detection are subject to WhatsApp's platform reliability and rate limits.
8. The LLM analysis pipeline depends on Google Gemini API and OpenAI API availability. Service outages, rate limit enforcement or pricing changes may temporarily interrupt the AI property analysis and trend recommendation features.
9. The lead scoring model is trained on Pakistani real estate listing data available at development time. Its accuracy will degrade as market conditions change and will require periodic retraining.
10. The system does not verify that a seller legally owns the property they are selling. Sellers are fully responsible for the accuracy and legal compliance of the information they submit through the property form.
11. Deal closure and revenue tracking are recorded manually by the user. LEADS does not verify that a deal actually closed or that the recorded commission amount is accurate.

# 9. Data Gathering Approach

Requirements for LEADS were gathered through four approaches. A structured technical analysis of four existing Pakistani real estate platforms Zameen.com, OLX Pakistan, Graana.com, Ilaan.com and was conducted to map their publicly accessible data fields, confirm scraping feasibility using Scrapy and Playwright, verify the absence of public comment sections or buyer behavior data and document each platform's anti-scraping measures, with findings directly shaping the scraper architecture and the decision to use on-platform buyer behavior rather than externally scraped intent signals. Unstructured interviews were conducted with real estate agents and property investors from Islamabad and Rawalpindi, focusing on how they currently identify investment opportunities, how long it takes to collect complete property details from sellers, how many messages they send before receiving a serious response and whether an AI-drafted outreach tool would be adopted in their workflow, with responses consistently identifying manual data collection and the inability to identify trending categories as their biggest time costs. A questionnaire was distributed among property buyers asking how they discover properties, what information is most frequently missing from listings they encounter and whether a platform with AI-enhanced verified listings would be preferred over existing options, with the majority citing incomplete listing information and fake listings as their primary frustrations. A technical feasibility study was conducted for each of the five scraping targets to confirm that category, price, location and seller contact data is publicly accessible, to identify which platforms require Playwright versus Scrapy and to validate the multi-agent parallel architecture as both technically sound and practically achievable within the project timeline..

# 10. Project Contribution

LEADS introduces several technical and conceptual contributions which do not exist in any current Pakistani real estate platform:

- **Computed Multi-Platform Market Intelligence:** LEADS is the first Pakistani platform to compute real-time market demand scores for property categories and cities by continuously analysing listing velocity, average days on market and price movement across five competing platforms simultaneously. No existing platform provides this intelligence to users.

- **Multi-Agent Parallel Scraping Architecture:** Rather than running a single scraper that can be blocked and takes hours to complete, LEADS launches one independent Celery worker per platform, each on a separate proxy IP, each with a configurable record limit, all running in

parallel. This multi-agent architecture collects targeted, filtered data from five platforms in minutes and is far more resilient to IP blocking than any single-agent approach.

- **Human-in-the-Loop AI Outreach Drafting:** LEADS uses LLM APIs to draft WhatsApp outreach messages, terms-and-conditions summaries and buyer campaign messages, all subject to mandatory user review before dispatch. This human-in-the-loop approach to AI-assisted outreach is not present in any Pakistani real estate tool.

- **Category-Specific Structured Property Data Collection:** LEADS delivers category-specific property data collection forms to sellers via WhatsApp, with different mandatory fields per property type. This replaces the back-and-forth messaging that currently dominates an agent's working day with a single structured submission that captures all required data at once.

- **LLM-Driven Property Valuation and Description Generation:** Submitted form data is passed to LLM APIs to generate a fair market valuation with minimum, maximum and recommended prices, identify hot selling features, flag issues, verify property category and produce an AI-enhanced listing description. This end-to-end property intelligence generation from structured seller input is novel in the Pakistani real estate context.

- **Webhook-Based Warm Lead Detection and Classification:** LEADS monitors all WhatsApp replies via the Cloud API webhook, classifies each reply as HOT, WARM or COLD based on content analysis and routes warm leads to the dashboard with an auto-drafted follow-up for user review. This automated warm lead detection from WhatsApp campaign replies is not available in any existing Pakistani real estate tool.

- **Dual-Channel OTP via WhatsApp:** All users must verify both email address and phone number via OTP before accessing any platform feature. Phone OTP is delivered via WhatsApp rather than SMS, reflecting the reality that WhatsApp penetration in Pakistan is significantly higher than SMS reliability.

- **Role-Based Three-Portal Architecture:** LEADS is constructed on the basis of three portals Seller and Agent Portal, Buyer Portal and Admin Portal. All three portals have completely different interface, feature set and view of data tailored exactly to the user using that portal.

**Contribution Impact**

Through integration of AI-driven listing enhancement and enrichment, lead scoring by means of NLP techniques, automation of multi-channel lead outreach, verification of user identities, gathering of live intelligence about the market and role-based portal architecture, LEADS changes the game in Pakistani real estate sector significantly. LEADS will allow sellers to leverage a real estate platform to their full potential. Rather than relying on traditional platforms where sellers just list their property and wait to sell it, LEADS will become a complete active seller-centric platform. With these contributions made, LEADS will be an innovative product indeed.

# 11. Relevance to Course Modules

LEADS is designed on the academic knowledge and technical skills gained throughout the Computer Science degree program at COMSATS University Islamabad. The following course modules are directly reflected in the design and development of the system:

- **Software Engineering:** Applied SDLC phases, requirement gathering via platform analysis and system design through architecture diagrams and module based WBS.

- **Object Oriented Programming (OOP):** Backend modules like scrapers, scoring algorithm, AI enhancer etc. all created in form of different classes in compliance with OOP standards of modularity and reusability.
- **Data Structures & Algorithms (DSA):** This is used in the lead scoring pipeline where the scraped information is sorted, ranked and filtered by applying weighted ranking algorithm.
- **Artificial Intelligence (AI):** Google Gemini is used for automatic listings enhancement and NLP-based classification of properties into categories before being posted.
- **Machine Learning (ML):** Development of supervised ML model for lead scoring through feature engineering, model selection and cross-validation on Pakistani real estate dataset.
- **Information Security:** Dual OTP authentication is used for all users along with secure credentials storage and rate limiting on login and OTP attempts to block brute force attacks.
- **Database Systems:** Dual database schema consisting of PostgreSQL for relational data and PostgreSQL for non-relational scraped data. Indexing and query optimization would be done.
- **Advanced Web Technologies**: Frontend development with React.js and Next.js, incorporation of RESTful API and server-side rendering for better search engine indexing.

# 12. Tools and Technologies

This section lists all the tools, technologies, programming languages and APIs that will be used to build and run LEADS. Each item is selected based on what it does best for that specific part of the system.

**Table 2: Tools and Technologies for LEADS**

| | Tool / Technologies | Version | Rationale |
|---|---|---|---|
| **Tools And Technologies** | VS Code | Latest | Primary code editor for both frontend and backend development |
| | Postman | Latest | API testing and endpoint validation during development |
| | Docker | 27+ | Keeps development and deployment environments consistent across machines |
| | pgAdmin | Latest | Visual interface for managing PostgreSQL database |
| | Git | Latest | Version control and team collaboration throughout development |
| | React.js | 19.2 | Component-based structure for building a modular and maintainable user interface |
| | Next.js | 16+ | Server-side rendering so listing pages can be indexed by Google search |
| | Tailwind CSS | 4+ | Fast and consistent styling across all pages without writing custom CSS |
| | Python | 3.13+ | Primary language for scraping pipeline, ML model and AI integrations |

| | | | |
|---|---|---|---|
| | FastAPI | 0.155+ | Handles concurrent scraping and AI tasks efficiently due to its async design |
| | Django + DRF | 5.1+ / 3.15+ | Provides built-in admin interface and clean management of relational data |
| | PostgreSQL | 17+ | Stores users, listings, leads, chat history and analytics with full reliability |
| | Scrapy | 2.12+ | Production-grade scraping framework with built-in proxy and rate limiting support |
| | Playwright | 1.50+ | Handles JavaScript-rendered pages that Scrapy alone cannot process |
| | Celery | 5.4+ | Manages scraping jobs and outreach tasks as background processes |
| | Redis | 7.4+ | Message broker for Celery task queue and application-level caching |
| | Google Gemini API | 2.0 Flash | Generates improved listing descriptions through structured prompts before publication |
| | OpenAI API | GPT-4o mini | Used for complex NLP tasks and buyer intent analysis in the scoring pipeline |
| | Hugging Face Transformers | 4.47+ | Pre-trained models for buyer intent scoring and property category classification |
| | scikit-learn | 1.6+ | Building, training and running the composite lead scoring model |
| | Meilisearch | 1.5+ | Fast open-source search engine for buyer listing discovery and filtering |
| | WhatsApp Cloud API | v21+ | Sending OTP verification messages and automated lead outreach via WhatsApp |
| | Resend | Latest | Transactional email delivery for OTP verification and lead outreach emails |
| | SMS Gateway | | Fallback outreach channel for leads who do not have WhatsApp |
| | Google Maps API | Latest | Location selection during listing creation and map display on listing detail pages |
| | JavaScript / TypeScript | EES2025 / 5.7+ | Frontend development and Next.js API routes |

# 13. Project Stakeholders and Roles

This section identifies all the people and groups who are involved in or affected by LEADS, along with their roles and responsibilities in the project.

**Table 3 Project Stakeholders for LEADS**

| Role | Details |
|---|---|
| **Project Sponsor** | COMSATS University Islamabad, Islamabad Campus |
| **Development Team** | Mamoon Tahir (FA23-BSE-051)<br>Muhammad Soban (FA23-BSE-056) |
| **Project Supervisor** | Dr. Tehseen Riaz Abbasi |
| **FYP Committee** | COMSATS FYP Evaluation Committee |

# 14. Module based Work Division

The following table outlines how the project modules and development responsibilities are divided between the two team members of this project.

**Table 4 Team Member Work Division for LEADS**

| Student Name | Student Registration Number | Responsibility/ Module / Feature |
|---|---|---|
| Mamoon Tahir | FA23-BSE-051 | Module 4<br>Module 5<br>Module 6<br>Module 8<br>Module 9<br>Module 12 |
| Muhammad Soban | FA23-BSE-056 | Module 1<br>Module 2<br>Module 3<br>Module 7<br>Module 10<br>Module 11 |

# 15. WBS and Gantt Chart

The Work Breakdown Structure (WBS) and Gantt Chart below provide a detailed timeline for the development of LEADS, covering all phases from scope and planning through to final delivery. All tasks are expanded to level 2, with estimated start and end dates, assigned resources and dependencies clearly identified for each task.

## WBS

**Table 5 WBS for LEADS**

| ID | Task Name | Start | Finish | Resource Names |
|---|---|---|---|---|
| **1** | **SCOPE AND PLANNING** | | | |
| 1.1 | Competitor Analysis | Mon 4/27/26 | Wed 4/29/26 | Mamoon Tahir, Muhammad Soban |
| 1.2 | Requirements Gathering | Thu 4/30/26 | Thu 5/7/26 | Mamoon Tahir, Muhammad Soban |
| 1.3 | Scope Document Writing | Fri 5/8/26 | Wed 5/13/26 | Mamoon Tahir, Muhammad Soban |
| 1.4 | Scope Submitted | Wed 5/13/26 | Wed 5/13/26 | Dr. Tehseen Riaz Abbasi, Mamoon Tahir, Muhammad Soban |
| **2** | **SYSTEM DESIGN** | | | |
| 2.1 | Database Schema Design | Thu 5/14/26 | Fri 5/22/26 | Muhammad Soban |
| 2.2 | System Architecture and API design | Mon 5/25/26 | Tue 6/2/26 | Mamoon Tahir, Muhammad Soban |
| 2.3 | Lead-scoring model and NLP reply-classification architecture planning | Wed 6/3/26 | Tue 6/9/26 | Mamoon Tahir |
| 2.4 | UI/UX wireframes | Wed 6/10/26 | Tue 6/23/26 | Mamoon Tahir, Muhammad Soban |
| 2.5 | Design Phase Complete | Tue 6/23/26 | Tue 6/23/26 | Dr. Tehseen Riaz Abbasi, Mamoon Tahir, Muhammad Soban |
| **3** | **SOFTWARE REQUIREMENT SPECIFICATION (SRS)** | | | |
| 3.1 | Functional Requirements Documentation | Wed 6/24/26 | Tue 6/30/26 | Mamoon Tahir, Muhammad Soban |
| 3.2 | Non Functional Requirements Documentation | Wed 7/1/26 | Fri 7/3/26 | Mamoon Tahir, Muhammad Soban |
| 3.3 | Usecase Diagrams and Descriptions | Mon 7/6/26 | Tue 7/14/26 | Mamoon Tahir, Muhammad Soban |
| 3.4 | UML Diagrams | Wed 7/15/26 | Thu 7/23/26 | Mamoon Tahir, Muhammad Soban |
| 3.5 | SRS Review and Finalization | Fri 7/24/26 | Mon 7/27/26 | Dr. Tehseen Riaz Abbasi, Mamoon Tahir, Muhammad Soban |
| 3.6 | SRS Complete | Mon 7/27/26 | Mon 7/27/26 | Dr. Tehseen Riaz Abbasi |

| ID | Task Name | Start | Finish | Resource Names |
|---|---|---|---|---|
| **4** | **SOFTWARE DESIGN SPECIFICATION (SDS)** | | | |
| 4.1 | Architecture and Component Design Documentation | Tue 7/28/26 | Thu 7/30/26 | Mamoon Tahir, Muhammad Soban |
| 4.2 | ER Diagram and Design Documentation | Fri 7/31/26 | Wed 8/11/26 | Mamoon Tahir, Muhammad Soban |
| 4.3 | UML Diagrams | Wed 8/12/26 | Wed 8/18/26 | Mamoon Tahir, Muhammad Soban |
| 4.4 | SDS Review and Finalization | Wed 8/19/26 | Thu 8/20/26 | Dr. Tehseen Riaz Abbasi, Mamoon Tahir, Muhammad Soban |
| 4.5 | SDS Complete | Thu 8/20/26 | Thu 8/20/26 | Dr. Tehseen Riaz Abbasi |
| **5** | **DEVELOPMENT** | | | |
| 5.1 | Environment Setup | Fri 8/21/26 | Mon 8/24/26 | Muhammad Soban |
| 5.2 | M01 - User Authentication | Tue 8/25/26 | Wed 9/2/26 | Muhammad Soban |
| 5.3 | M02 - Market Intelligence Engine | Thu 9/3/26 | Mon 9/14/26 | Muhammad Soban |
| 5.4 | M03 - Scraping Engine - Scrapy Spiders | Tue 9/15/26 | Thu 9/24/26 | Muhammad Soban |
| 5.5 | M03 - Scraping Engine - Playwright Spiders | Fri 9/25/26 | Tue 10/6/26 | Muhammad Soban |
| 5.6 | Scraping and Market Intelligence Complete | Tue 10/6/26 | Tue 10/6/26 | Dr. Tehseen Riaz Abbasi |
| 5.7 | M04 - Seller Outreach Agent | Wed 10/7/26 | Thu 10/15/26 | Mamoon Tahir |
| 5.8 | M05 - Seller Form System | Fri 10/16/26 | Mon 10/26/26 | Mamoon Tahir |
| 5.9 | M06 - AI Property Analyser (two stage LLM chain) | Tue 10/27/26 | Thu 11/5/26 | Mamoon Tahir |
| 5.10 | M07 Buyer Outreach - Lead Scoring model (feature engineering, scikit-learn training, cross validation) | Fri 11/6/26 | Thu 11/19/26 | Mamoon Tahir |
| 5.11 | M07 Buyer Outreach - Campaign dispatch + webhook reply NLP classification | Fri 11/20/26 | Thu 12/3/26 | Mamoon Tahir |
| 5.12 | Seller Side Pipeline Complete | Thu 12/3/26 | Thu 12/3/26 | Dr. Tehseen Riaz Abbasi |
| 5.13 | M08 - Listing Management System - publish pipeline, SSR listing pages, Meilisearch indexing | Fri 12/4/26 | Mon 12/14/26 | Muhammad Soban |
| 5.14 | M08 - Listing Management System - Buyer Discovery UI | Tue 12/15/26 | Wed 12/23/26 | Muhammad Soban |

| ID | Task Name | Start | Finish | Resource Names |
|---|---|---|---|---|
| | (filters, bookmarks, saved searches, contact routing) | | | |
| 5.15 | M09 - Seller Dashboard (deal/lead summary, activity chart, revenue tracking) | Thu 12/24/26 | Thu 12/31/26 | Mamoon Tahir |
| 5.16 | M10 - Admin Control Panel (scrapers, proxies, API keys, scoring weights, user management, audit log) | Fri 1/1/27 | Tue 1/12/27 | Muhammad Soban |
| 5.17 | M11 - Campaign Dashboard | Wed 1/13/27 | Wed 1/20/27 | Muhammad Soban |
| 5.18 | M12 - Profile and Settings | Thu 1/21/27 | Wed 1/27/27 | Mamoon Tahir |
| 5.19 | Core Modules Development Complete | Wed 1/27/27 | Wed 1/27/27 | Dr. Tehseen Riaz Abbasi |
| 5.20 | Integration of all 12 modules into a unified system | Thu 1/28/27 | Fri 2/5/27 | Mamoon Tahir, Muhammad Soban |
| 5.21 | End-to-end pipeline testing | Mon 2/8/27 | Thu 2/11/27 | Mamoon Tahir, Muhammad Soban |
| 5.22 | 100% Working Product Complete | Thu 2/11/27 | Thu 2/11/27 | Dr. Tehseen Riaz Abbasi |
| **6** | **TESTING** | | | |
| 6.1 | Unit test cases for all modules | Fri 2/12/27 | Wed 2/17/27 | Mamoon Tahir, Muhammad Soban |
| 6.2 | Integration test plan (end-to-end pipeline) | Thu 2/18/27 | Tue 2/23/27 | Mamoon Tahir, Muhammad Soban |
| 6.3 | Test results documentation and traceability matrix | Wed 2/24/27 | Fri 2/26/27 | Mamoon Tahir, Muhammad Soban |
| 6.4 | Testing Document Complete | Fri 2/26/27 | Fri 2/26/27 | Dr. Tehseen Riaz Abbasi |
| **7** | **FINAL REVIEW AND DELIVERY** | | | |
| 7.1 | Lead Scoring and reply-classification model evaluation and retraining on live scraped data | Mon 3/1/27 | Fri 3/5/27 | Mamoon Tahir, Muhammad Soban |
| 7.2 | Performance optimization | Mon 3/8/27 | Fri 3/12/27 | Mamoon Tahir, Muhammad Soban |
| 7.3 | Security Review | Mon 3/15/27 | Fri 3/19/27 | Mamoon Tahir, Muhammad Soban |
| 7.4 | User Acceptance Testing | Mon 3/22/27 | Mon 3/29/27 | Mamoon Tahir, Muhammad Soban |
| 7.5 | Bug fixing and final polishing based on UAT feedback | Tue 3/30/27 | Fri 4/2/27 | Mamoon Tahir, Muhammad Soban |
| 7.6 | Final Report Writing | Mon 4/5/27 | Tue 4/13/27 | Mamoon Tahir, Muhammad Soban |
| 7.7 | Final Presentation Preparation | Wed 4/14/27 | Thu 4/22/27 | Dr. Tehseen Riaz Abbasi, Mamoon Tahir, Muhammad Soban |

| ID | Task Name | Start | Finish | Resource Names |
|---|---|---|---|---|
| 7.8 | Plagiarism Check | Fri 4/23/27 | Fri 4/23/27 | Mamoon Tahir, Muhammad Soban |
| 7.9 | Final Evaluation and Presentation | Mon 4/26/27 | Mon 4/26/27 | Mamoon Tahir, Muhammad Soban |
| 7.10 | Project Complete | Mon 4/26/27 | Mon 4/26/27 | Dr. Tehseen Riaz Abbasi, Mamoon Tahir, Muhammad Soban |

## Gantt Chart

The Gantt chart below provides a detailed timeline for the development of LEADS, covering all phases from scope and planning through to final delivery. It is broken into three separate parts for better visibility.

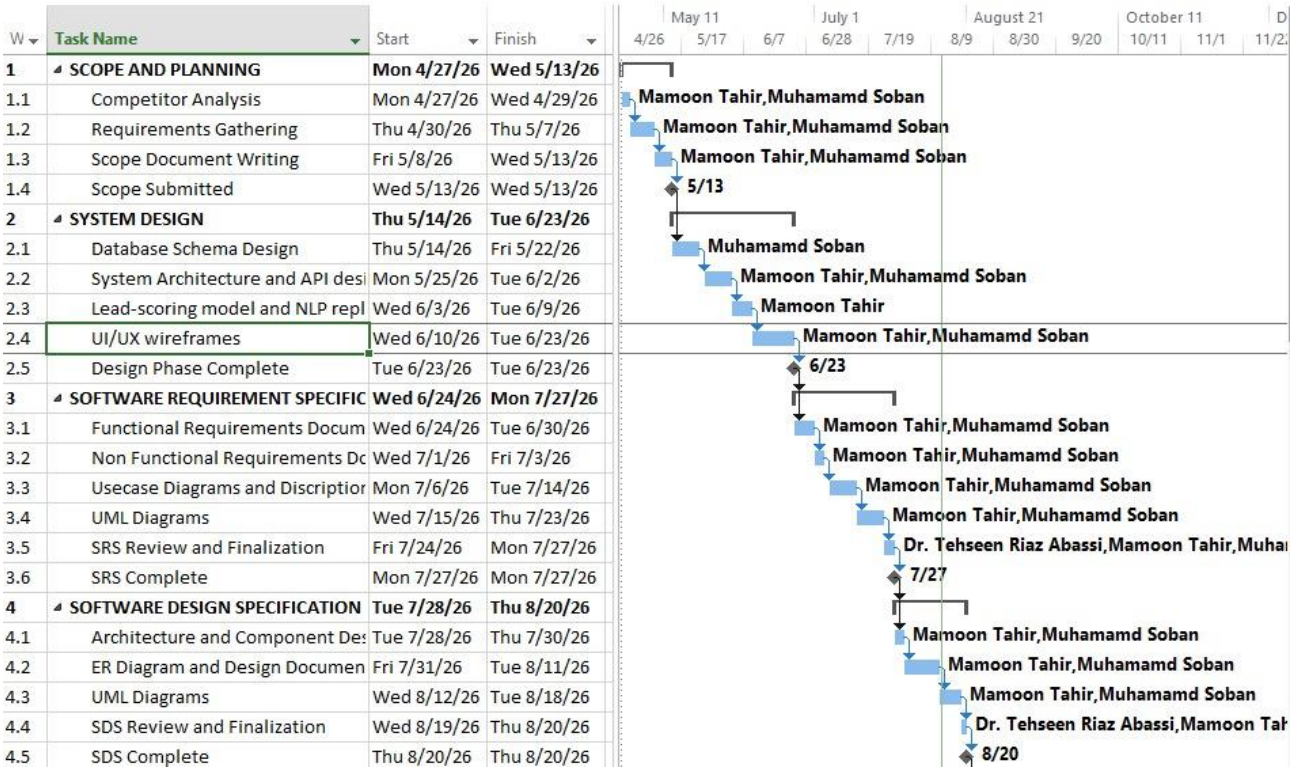
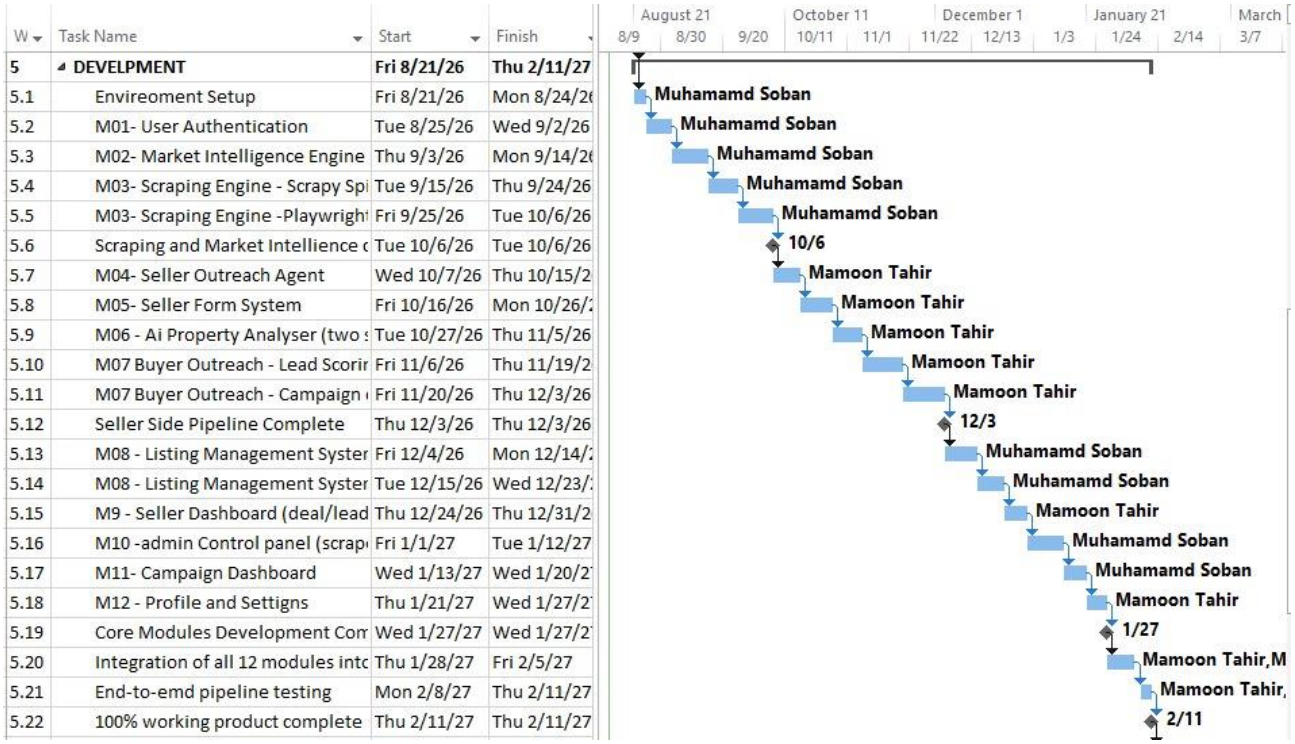



**Figure 15.1: Gantt Chart part 1 (Scope, Planning and System Design)**



| W | Task Name | Start | Finish |
|---|---|---|---|
| 5 | **DEVELPMENT** | **Fri 8/21/26** | **Thu 2/11/27** |
| 5.1 | Envireoment Setup | Fri 8/21/26 | Mon 8/24/2 |
| 5.2 | M01- User Authentication | Tue 8/25/26 | Wed 9/2/26 |
| 5.3 | M02- Market Intelligence Engine | Thu 9/3/26 | Mon 9/14/2 |
| 5.4 | M03- Scraping Engine - Scrapy Spi | Tue 9/15/26 | Thu 9/24/26 |
| 5.5 | M03- Scraping Engine -Playwright | Fri 9/25/26 | Tue 10/6/26 |
| 5.6 | Scraping and Market Intellience c | Tue 10/6/26 | Tue 10/6/26 |
| 5.7 | M04- Seller Outreach Agent | Wed 10/7/26 | Thu 10/15/2 |
| 5.8 | M05- Seller Form System | Fri 10/16/26 | Mon 10/26/ |
| 5.9 | M06 - Ai Property Analyser (two s | Tue 10/27/26 | Thu 11/5/26 |
| 5.10 | M07 Buyer Outreach - Lead Scorir | Fri 11/6/26 | Thu 11/19/2 |
| 5.11 | M07 Buyer Outreach - Campaign ( | Fri 11/20/26 | Thu 12/3/26 |
| 5.12 | Seller Side Pipeline Complete | Thu 12/3/26 | Thu 12/3/26 |
| 5.13 | M08 - Listing Management Syster | Fri 12/4/26 | Mon 12/14/ |
| 5.14 | M08 - Listing Management Syster | Tue 12/15/26 | Wed 12/23/ |
| 5.15 | M9 - Seller Dashboard (deal/lead | Thu 12/24/26 | Thu 12/31/2 |
| 5.16 | M10 -admin Control panel (scrap | Fri 1/1/27 | Tue 1/12/27 |
| 5.17 | M11- Campaign Dashboard | Wed 1/13/27 | Wed 1/20/2 |
| 5.18 | M12 - Profile and Settigns | Thu 1/21/27 | Wed 1/27/2 |
| 5.19 | Core Modules Development Com | Wed 1/27/27 | Wed 1/27/2 |
| 5.20 | Integration of all 12 modules intc | Thu 1/28/27 | Fri 2/5/27 |
| 5.21 | End-to-emd pipeline testing | Mon 2/8/27 | Thu 2/11/27 |
| 5.22 | 100% working product complete | Thu 2/11/27 | Thu 2/11/27 |



**Figure 15.2: Gantt Chart part 2 (Development)**



| 6 | **TESTING** | **Fri 2/12/27** | **Fri 2/26/27** |
|---|---|---|---|
| 6.1 | Unit test cases for all modules | Fri 2/12/27 | Wed 2/17/2 |
| 6.2 | Integration test plan (end-end pipeline) | Thu 2/18/27 | Tue 2/23/27 |
| 6.3 | Test results documentation and tracibility matrix | Wed 2/24/27 | Fri 2/26/27 |
| 6.4 | Testing Document Complete | Fri 2/26/27 | Fri 2/26/27 |
| 7 | **FINAL REVIEW AND DELIVERY** | **Mon 3/1/27** | **Mon 4/26/2** |
| 7.1 | Lead Scoring and reply-classification modelevaluation and retraining on live scrapped data | Mon 3/1/27 | Fri 3/5/27 |
| 7.2 | Performance optimization | Mon 3/8/27 | Fri 3/12/27 |
| 7.3 | Security Review | Mon 3/15/27 | Fri 3/19/27 |
| 7.4 | User Acceptance Testing | Mon 3/22/27 | Mon 3/29/2 |
| 7.5 | Bug fixing and final polishing based on UAT feedback | Tue 3/30/27 | Fri 4/2/27 |
| 7.6 | Final Report Writing | Mon 4/5/27 | Tue 4/13/27 |
| 7.7 | Final Presentation preperation | Wed 4/14/27 | Thu 4/22/27 |
| 7.8 | Plagiarism Check | Fri 4/23/27 | Fri 4/23/27 |
| 7.9 | Final Evalutaion and Presentation | Mon 4/26/27 | Mon 4/26/2 |
| 7.10 | Project Complete | Mon 4/26/27 | Mon 4/26/2 |



**Figure 15.3: Gantt Chart part 3 (Testing, Final Review and Delivery**

# 16. Mockups

The following mockups represent the major modules of LEADS across all three actor's UI.

### Seller / Agent Dashboard

This screen displays the seller's active listings, lead count, outreach performance and recent buyer messages in one place.

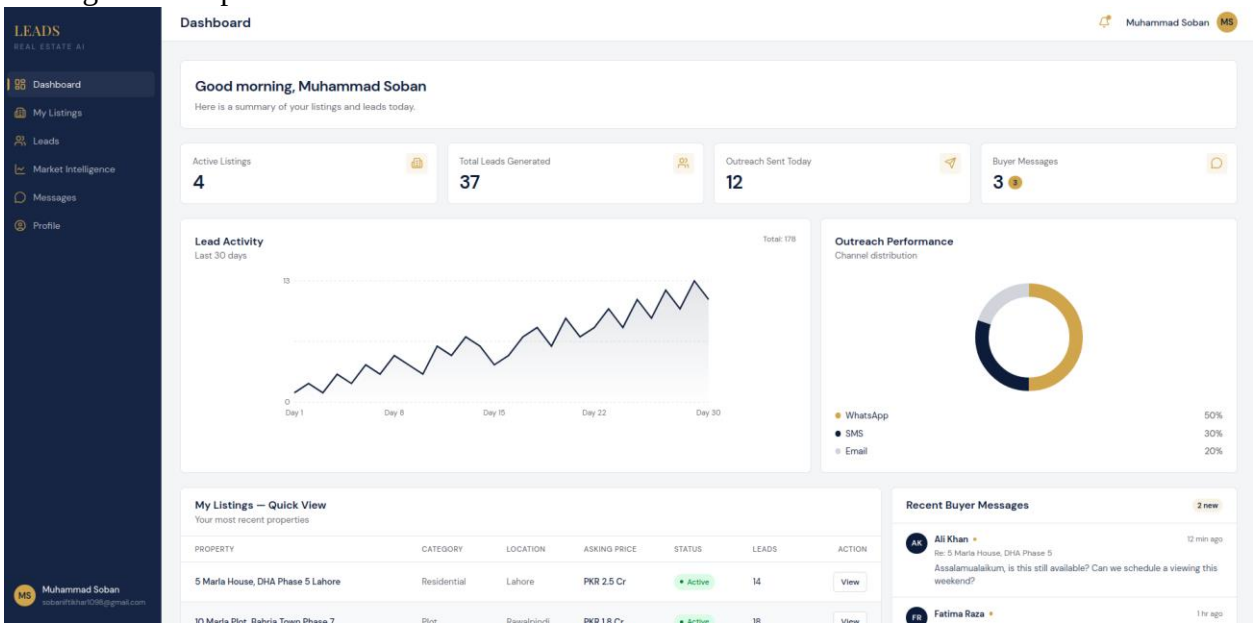



**Figure 16.1: Seller / Agent Dashboard**

### Leads Management

This screen shows the seller a ranked list of scored buyer leads with outreach status across WhatsApp, SMS and email for a specific listing.

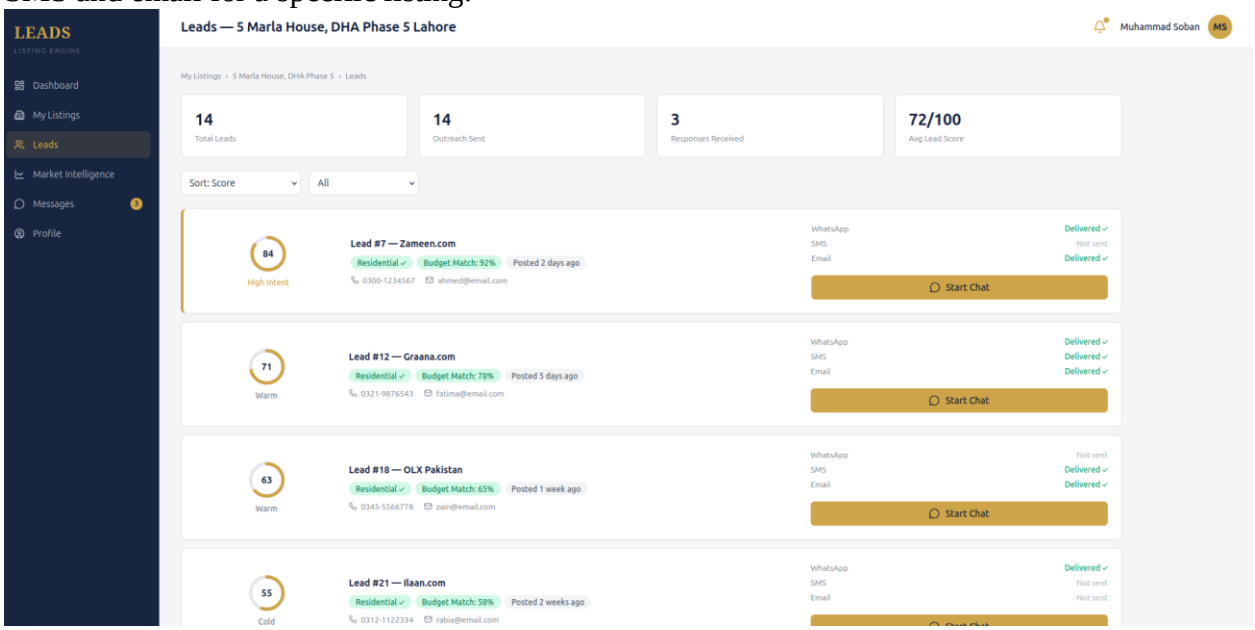



**Figure 16.2: Lead Management**

## Market Intelligence

This screen shows the seller a live price comparison against competitor platforms and a demand indicator for their property type and city.

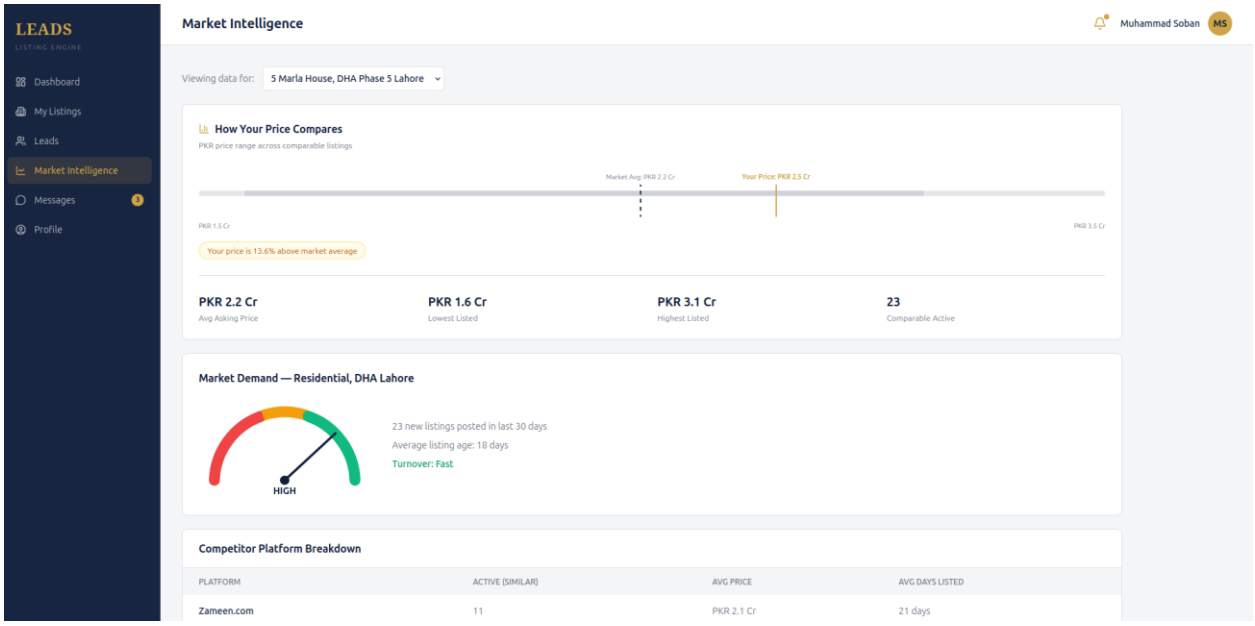



**Figure 16.3: Market Intelligence**

## Buyer Browse Listings

This screen allows buyers to browse, filter and save verified AI-reviewed property listings across Pakistan.

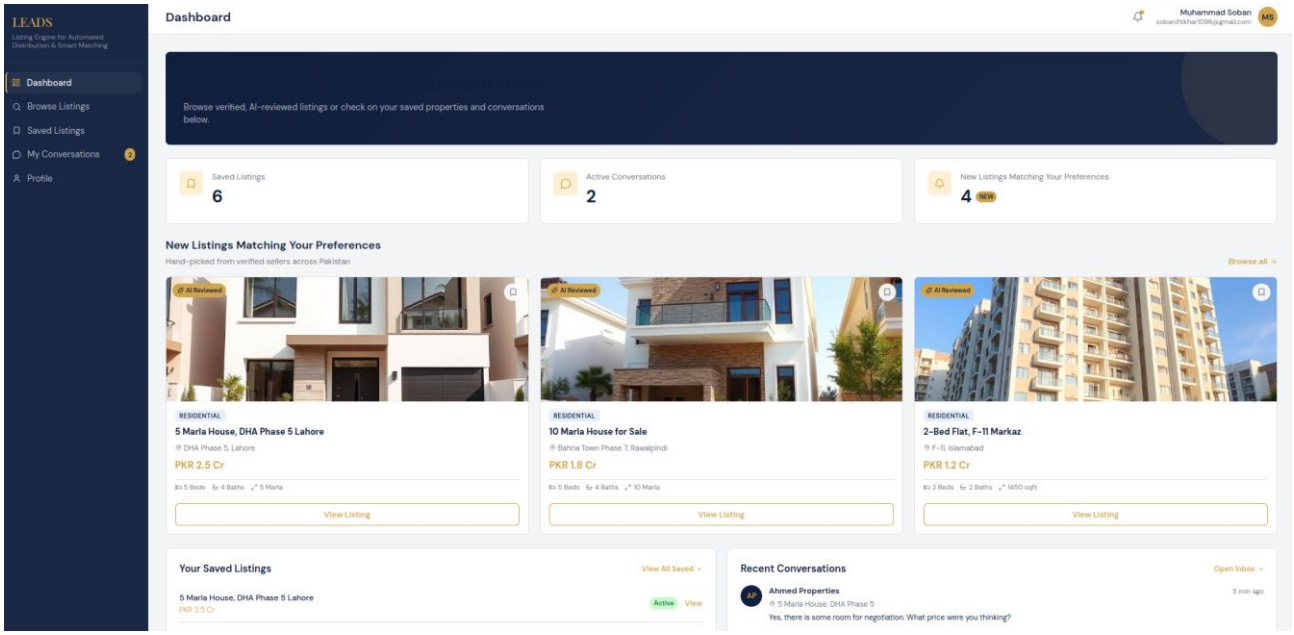



**Figure 16.4: Buyer Browse Listings**

## Admin Dashboard

This screen displays platform-wide statistics including total users, active listings, leads generated, scraper health and system uptime in one place.

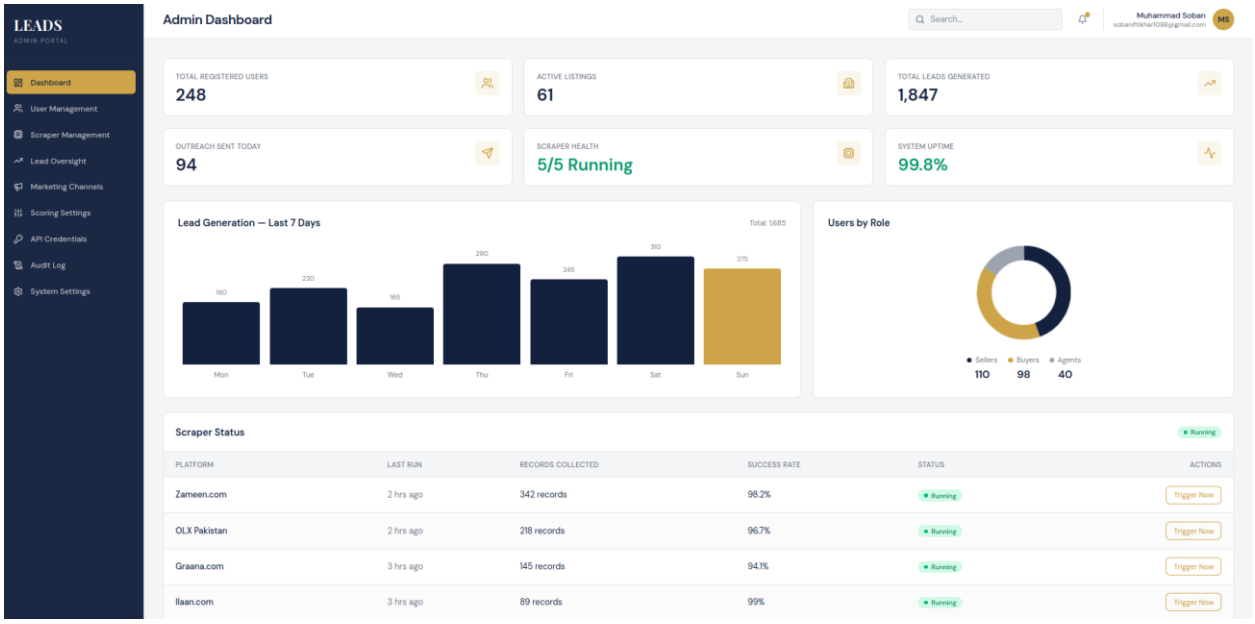



**Figure 16.5: Admin Dashboard**

## Scraper Management

This screen gives the administrator full control and visibility over all five platform scrapers including status, records collected and error logs.

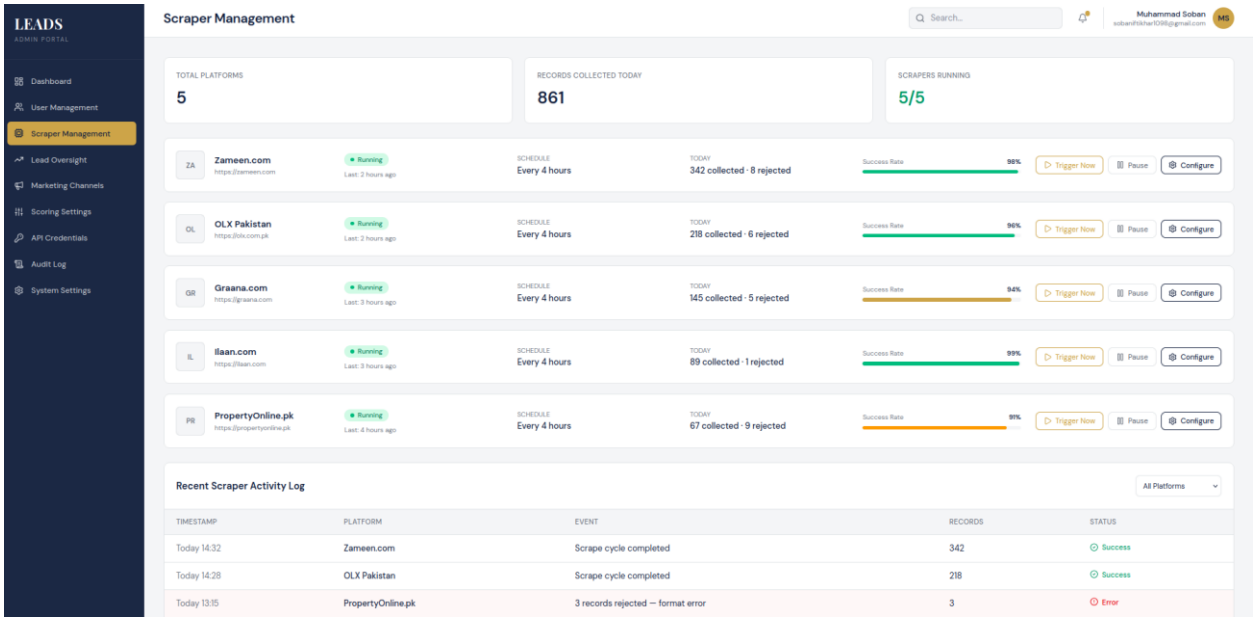



***Figure 16.6: Scraper Management***

# 17. References

The following books, research papers and web sources were used during the analysis, design and planning phases of LEADS.

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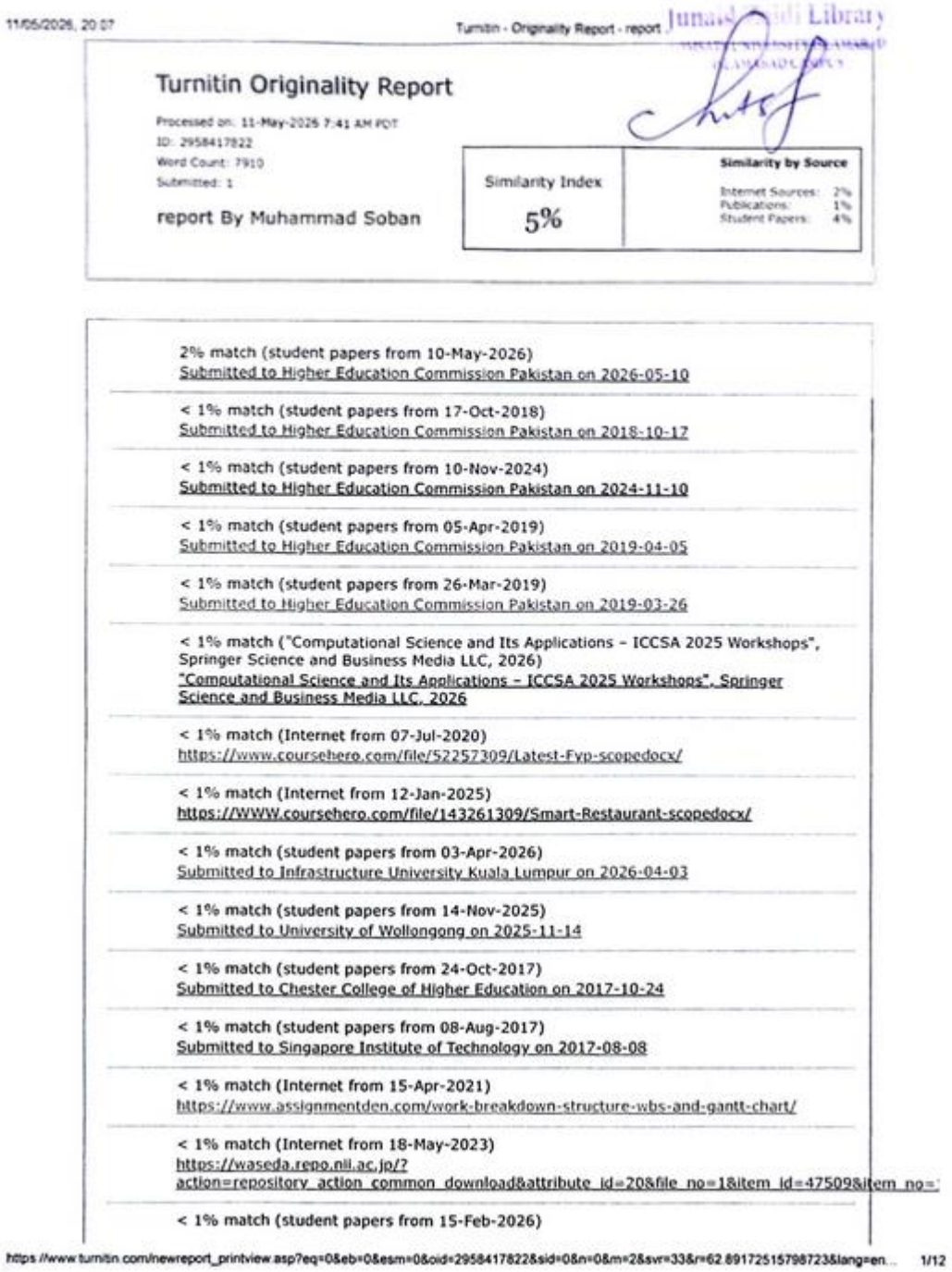
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# 18. Plagiarism Report



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